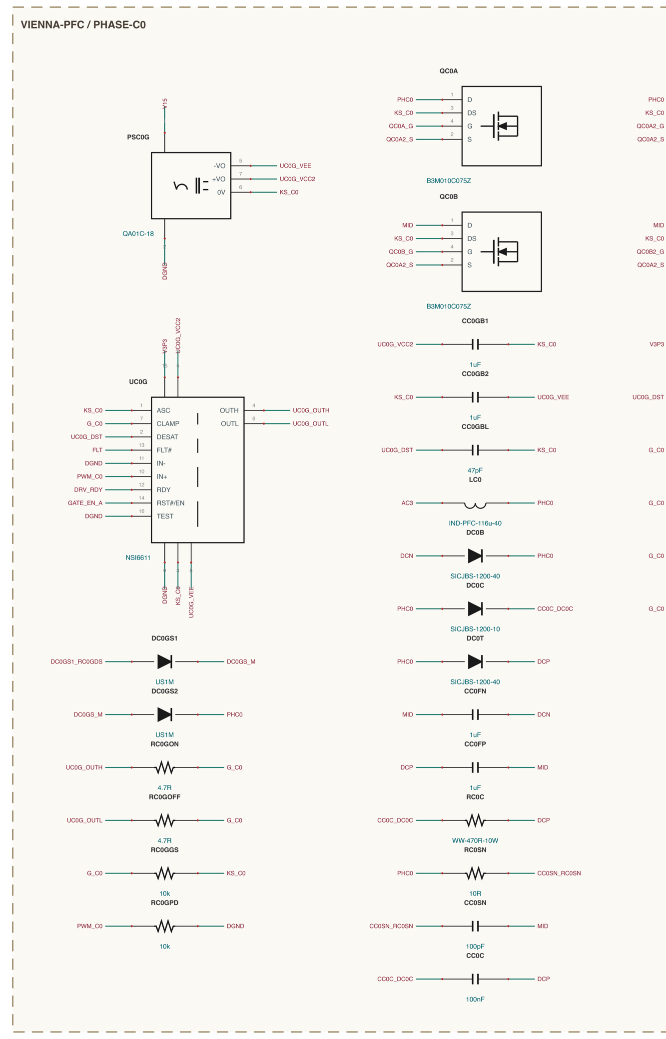




SHEET INDEX

40 kW AC-DC (lower) - 1 of 3

AC-SENSING - 9 sections

AUX-POWER - 5 sections

CONTROL - 4 sections

DC-LINK - 4 sections

INPUT-EMI - 4 sections

VIENNA-PFC - 3 sections

rev D.4 - 29 sections - 324 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D1 LA0-LC0: 5x 0077908A7 KoolMu (CATALOG AL37), N=26 +/-1 lot trim -> L0 116uH, >=61uH @104A pk (E51)

D6 LDM1-3: 2x T57 60u, N=8, 26.4mm2 -> 10.5uH @109A pk (floor 9.2uH)

total Lr = trim + measured xfmr leakage (bins COMPENSATE leakage spread); Cr at TANK sections

40 kW ACDC board 1of2 - Vienna PFC (1x hot cells)

DC-Modules 40 kW - board AC-DC (lower), sheet 1 of 3

2026-09-06 · rev D.4